



Predicting Adult Autism Screening Outcomes Using Logistic Regression

A Data-Driven Approach to Preliminary ASD Screening

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01.

INTRODUCTION



Why Use Data for Autism
Screening?



Why This Matters (Background + Context)



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- Autism Diagnoses are often delayed
- Early identification can improve access to support and accommodations
- Screening tools like the Autism Spectrum Quotient (AQ-10) offer a quick alternative

Characteristics Present → Waitlist → Long Assessment → Delayed Diagnosis

Vs

Quick Screen → Flagged for Evaluation → Earlier Support and Accommodations

Goal: Can we build a model that flags individuals who may need further evaluation?



How Does the AQ-10 Work?



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Taking the test

The AQ-10 consists of **10** statements, giving you **4** choices for each statement:

1. Definitely agree
2. Slightly agree
3. Slightly disagree
4. Definitely disagree

Note: it makes no difference to your score whether you choose slightly or definitely, so just treat the statements as a binary choice of 'agree' and 'disagree'.

(Embrace Autism)



Scoring

- Scoring range: **0–10**
- Threshold score: **6↑**
 - **6+** you might be autistic
 - **<6** you may not be autistic
- **80%** of people designated Asperger's syndrome score **6** or higher

The AQ-10 is only available as a self-scoring test.

(Embrace Autism)



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02.

MODELING APPROACH



How were the models built?



What's the Dataset



Dataset Name: *Autistic Spectrum Disorder Screening Data for Adults*

704 Observations with 21 Variables

A mix of numerical, binary, and categorical

- 10 AQ-10 Survey Items -> Scored 0 or 1
- Outcome: Diagnosed w/ ASD (Yes/No)
- Also includes: age, gender, jaundice at birth, ethnicity, country of residence, sum of score, etc.
- With some missing values and mistyped values



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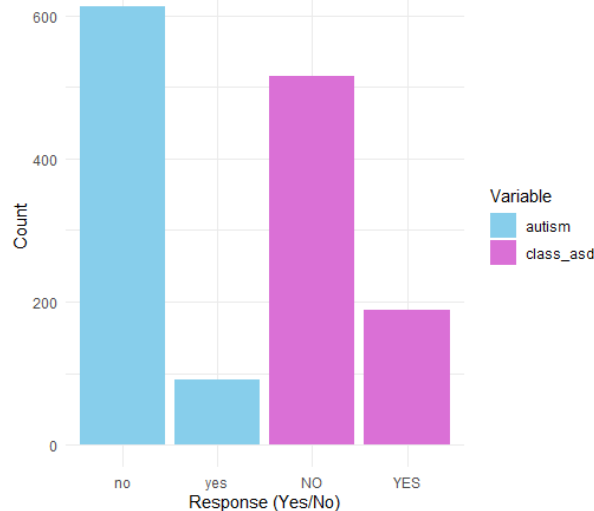
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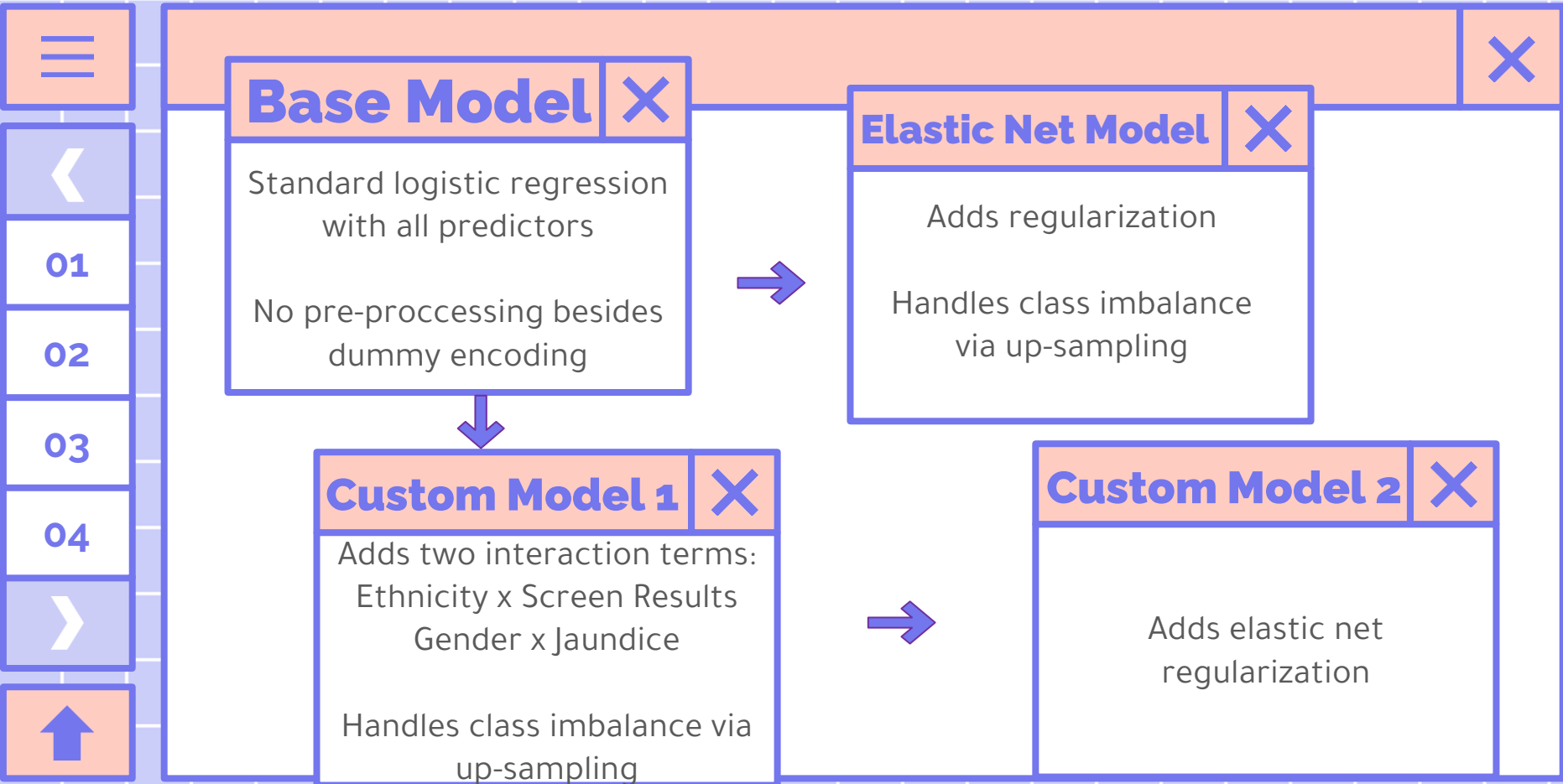
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Comparison of Autism Diagnosis and App Classification







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MODEL RESULTS



Which Model Possess the most
Potential?



Model Metrics



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	Accuracy	Brier	Kappa	AUC	Spec	Sens	X
BM	0.846	0.124	0.116	0.620	0.951	0.139	
ENM	0.680	0.202	0.187	0.708	0.686	0.640	
CM1	0.696	0.219	0.156	0.608	0.721	0.528	
CM2	0.683	0.201	0.183	0.711	0.692	0.625	



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CONCLUSION



What are some further
considerations?



Conclusion, Limitations, & Further Considerations



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Winning model: Custom Model 2

Why: Most of the model metrics for CM2 were better and the 2 strongest predictors being a survey question (Q4) and the interaction term of gender and jaundice.

Limitations: Small dataset, models don't have ideal sensitivity/specificity, important variables are "self-reported", issues with generalizability.

Suggestions: Bigger datasets, perhaps inclusion of additional surveys and survey data.



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THANKS!

Do you have any questions?

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